

MD SHARIFUL ALAM

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EDUCATION

New York Institute of Technology, Old Westbury, NY

Expected: December 2026

B.S. in Mechanical Engineering, Minor in Mathematics • GPA: 3.99/4.00

Honors: Presidential Honor List (7/7 semesters), CoECS Dean's Recognition Scholarship (2024, 2026), T.K. Steele Scholarship

Coursework: Machine Design, Heat Transfer, Fluid Mechanics, System Dynamics & Vibrations, Propulsion, Mechatronics

TECHNICAL SKILLS

- **FEA & Simulation:** Abaqus/CAE, Altair HyperMesh, MATLAB/Simulink
- **CAD & Mechanical Design:** SolidWorks, Autodesk Inventor, Fusion 360, AutoCAD
- **Programming & Embedded:** MATLAB, Arduino C/C++, C#
- **Engineering Tools:** Ansys GRANTA EduPack, Multisim, Blender
- **Manufacturing & Prototyping:** Manual Milling, Lathe, CNC Machining, 3D Printing

RESEARCH EXPERIENCE

Research Assistant — Dr. Wei Zeng, NYIT

May 2026 – Present

- Develop stochastic trabecular bone geometries and nonlinear bone-screw pullout models using MATLAB, Blender, HyperMesh, and Abaqus/CAE; configure meshing, material properties, contact/friction, and displacement-controlled axial screw-pullout loading conditions.
- Compare trabecular porosity, cortical shell presence, and screw geometry effects on pullout response while troubleshooting mesh quality and nonlinear convergence.

Research Assistant — Dr. Tindaro Ioppolo, NYIT

September 2025 – April 2026

- Collaborated with a faculty researcher under NYIT's Undergraduate Research and Entrepreneurship Program (UREP) to improve an optical microsphere-based wall shear stress sensor using whispering gallery mode (WGM) resonance.
- Redesigned sensor geometry in SolidWorks to reduce membrane stiffness and improve deformation transfer; supported microsphere fabrication and sensor assembly for future experimental validation.

PROJECTS

Half-Car Suspension System Modeling and Simulation — NYIT

Fall 2025

- Developed a 4-DOF suspension model in MATLAB from first principles; evaluated four parameter sets to meet 2 of 3 ride-comfort targets, reducing pitch rate from 0.57 to 0.50 rad/s and settling time from 3.69 to 2.82 s.

Autonomous Multi-Mode Vehicle with IMU Guidance — NYIT

Spring 2026

- Designed and built a 3D-printed differential-drive vehicle in SolidWorks, integrating Arduino C/C++ control, IMU guidance, a servo-mounted ultrasonic sensor, and custom circuitry for obstacle avoidance and human-following modes.

Countershaft Machine Element Design & Fatigue Analysis — NYIT

Fall 2025

- Analyzed a multi-pulley countershaft by calculating belt forces, bearing reactions, bending moments, and torque, then applied stress concentrations, endurance corrections, and the DE-Soderberg criterion with a factor of safety of 3 to size the shaft for infinite-life fatigue design.

EXPERIENCE

Peer Tutor — NYIT Office of Academic Success and Enrichment, Old Westbury, NY

March 2024 – Present

- Tutor undergraduate students in manufacturing, thermodynamics, CAD, MATLAB, physics, and circuits, adapting instruction to individual learning needs and recurring problem areas.

Undergraduate Math Tutor — NYIT Math Resource Center, Old Westbury, NY

October 2025 – Present

- Tutor students in Precalculus, Differential Equations, and Linear Algebra, providing individualized problem-solving support and tracking student progress.

Accounting Assistant — Hasan Tax & Accounting Inc., Elmhurst, NY

March 2024 – Present

- Support tax preparation, payroll processing, bookkeeping, and maintenance of confidential financial and client records.